

A Composable Four-Gate DevSecOps Pipeline for Python: Integrating SAST, Secret Detection, AST Analysis, and Dependency Auditing

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Abstract—Modern Python applications face a diverse threat landscape encompassing injection vulnerabilities, inadvertently committed secrets, weak cryptographic primitives, and transitively vulnerable dependencies. Existing solutions address each concern in isolation; developers must assemble ad-hoc pipelines whose gate interactions are poorly understood. We present *secure-python-pipeline*, a composable, open-source DevSecOps template that chains four automated security gates—Semgrep (SAST), Trufflehog v3 (secret detection), Bandit (Python AST analysis), and pip-audit (dependency CVE auditing)—into a reproducible GitHub Actions workflow. We evaluate the pipeline against five widely-adopted OSS Python projects totalling over 230 000 GitHub stars (*httpie*, *rich*, *FastAPI*, *black*, *ruff*) and demonstrate that three of the five pass all gates at their latest stable release, while two exhibit real, previously-undocumented findings that we characterise and report. The template ships with eight CWE-mapped custom Semgrep rules, a reference FastAPI application demonstrating correct mitigations, and a local-execution script that mirrors the CI environment exactly.

Index Terms—DevSecOps, SAST, Python security, Semgrep, Bandit, secrets detection, dependency auditing, CI/CD, OWASP Top 10, CWE

I. INTRODUCTION

Software supply-chain attacks increased 742% between 2019 and 2022 [1]. For Python specifically, the PyPI ecosystem hosts over 500 000 packages, and any one of them may introduce a transitive dependency carrying a known CVE. Simultaneously, OWASP’s 2021 Top Ten [2] continues to list injection (A03) and identification failures (A07) among the most prevalent web-application risks—vulnerabilities that are, in principle, detectable through static analysis before deployment.

Despite the availability of individual tools such as Bandit [3], Semgrep [4], Trufflehog [5], and pip-audit [6], no reference architecture exists that: (a) chains all four into a single actionable workflow; (b) documents the interaction semantics between gates; (c) provides a correctly-secured reference application that passes all gates; and (d) evaluates the pipeline empirically against real-world OSS code.

This paper makes the following contributions:

- 1) A four-gate pipeline architecture with documented gate ordering rationale and failure semantics (§IV).
- 2) Eight CWE-mapped Semgrep rules targeting Python-specific OWASP Top 10 patterns (§VI).
- 3) An empirical evaluation across five high-profile OSS projects (§VIII).
- 4) A reference FastAPI application demonstrating correct mitigations (§VII).

II. BACKGROUND AND RELATED WORK

A. Static Application Security Testing

SAST tools analyse source code without executing it. Semgrep [4] expresses rules as structural patterns over an AST, achieving near-zero false-negative rates on syntactically exact matches. Bandit [3] applies Python-specific heuristics (e.g., subprocess with `shell=True`, use of `pickle`) via an AST visitor, reporting severity and confidence independently.

B. Secret Detection

Trufflehog v3 [5] scans git history using entropy heuristics and over 800 regex-based detectors, then attempts live validation to confirm that a candidate secret is currently active. Only *verified* secrets fail the pipeline by default, dramatically reducing false-positive noise.

C. Software Composition Analysis

pip-audit [6] queries the Python Packaging Advisory Database (PyPA) and the OSV database for CVEs affecting pinned dependencies. Unlike *safety*, it accepts a `requirements.txt` without requiring installation of the packages, making it suitable for CI environments.

D. Prior Work

Pashchenko et al. [7] study vulnerable OSS dependencies in npm; Decan et al. [8] extend this to PyPI, finding that 11% of packages transitively depend on a known-vulnerable release. Our work complements this with an automated, actionable pipeline rather than a retrospective analysis. Zahan et al. [9] propose a threat model for supply-chain attacks; our pipeline operationalises their recommended controls at the CI layer.

III. THREAT MODEL

The pipeline targets the following threat categories within a Python project’s continuous integration environment:

T1 – Source-level injection: An attacker (or negligent developer) introduces code that constructs SQL queries, shell commands, or HTTP requests from unvalidated input. Semgrep’s structural patterns detect these at the AST level before the code reaches production.

T2 – Credential leakage: A secret (API key, private key, database password) is committed to version control, either in the current working tree or in a past commit that has since been reverted. Trufflehog scans the *entire* git history with `fetch-depth: 0` and suppresses unconfirmed candidates using live verification.

T3 – Insecure cryptographic primitives: Passwords hashed with MD5 or SHA-1 can be inverted at \$0.001 per hash on modern GPU clusters. Bandit and custom Semgrep rules flag these at ERROR severity.

T4 – Vulnerable transitive dependencies: A dependency introduced for unrelated functionality may carry a CVE that is exploitable in the project’s deployment context. pip-audit resolves the full dependency graph from `requirements*.txt` and checks each pinned version against the PyPA and OSV advisory databases.

Out of scope: Runtime exploitation (memory corruption, race conditions), social-engineering attacks on the developer, and adversarial model inputs in ML pipelines are explicitly excluded from this work. The pipeline operates purely on static artefacts.

IV. PIPELINE DESIGN

A. Gate Ordering

The four gates are ordered by *cost of failure*: gates that produce the most developer-actionable signal earliest are placed first.

- 1) **Gate 1 – Semgrep (SAST):** Structural AST matching catches injection sinks, insecure deserialisation, and debug flags in $O(n)$ time. Runs in a pinned container (`returntocorp/semgrep:1.96.0`) for reproducibility.
- 2) **Gate 2 – Trufflehog v3 (secrets):** Scans the full git history (`fetch-depth: 0`) using only the `--only-verified` flag to suppress unconfirmed candidates.
- 3) **Gate 3 – Bandit (Python AST):** Complements Semgrep with Python-semantic checks that require type information (e.g., SSL context verification disabled, weak hash algorithm selection).
- 4) **Gate 4 – pip-audit (dependencies):** Audits every `requirements*.txt` against PyPA advisory data without requiring a live installation.

Gates 1–4 run in parallel in CI (needs graph in GitHub Actions), converging on a summary job that posts a gate-result table to the job summary.

B. Failure Semantics

All ERROR-severity findings fail the pipeline immediately. WARNING-severity Semgrep findings (e.g., SSRF risk) are surfaced as annotations but do not block merge, allowing teams to triage asynchronously. This two-tier approach follows the principle of *graduated enforcement* [10].

C. Configuration and Reproducibility

Each gate is pinned to an exact tool version in the workflow file (`bandit==1.8.6`, `pip-audit==2.7.3`, `returntocorp/semgrep:1.96.0`) to ensure that CI results are reproducible across machines and over time. The Semgrep container image is pinned by tag rather than digest to remain practical while still preventing silent updates.

The `bandit.yml` configuration excludes `eval/cloned_repos/` (vendored OSS code) and `.venv/` to avoid noise from third-party artefacts. Trufflehog’s `config.yml` similarly excludes `example-project/tests/fixtures/` which contains intentionally fake secrets used by unit tests.

D. Local Execution

The `scripts/local_scan.sh` script mirrors the CI gates exactly, using the same tool versions and configuration files. It produces coloured pass/fail output and writes JSON reports to `.scan-reports/` for IDE integration. Developers can run the full pipeline locally in under two minutes on a typical codebase, catching issues before pushing.

E. PR Check Workflow

A lightweight secondary workflow (`pr-check.yml`) runs on every pull request and applies Bandit and pip-audit only to files changed in the PR diff. This reduces median CI time from ≈ 4 min (full scan) to ≈ 45 s, preserving developer feedback latency without sacrificing correctness.

The workflow uses GitHub Actions’ `concurrency` feature with `cancel-in-progress: true` so that a rapid sequence of commits to the same PR branch does not queue multiple redundant scans, preventing resource waste on low-powered self-hosted runners.

V. IMPLEMENTATION

A. Repository Layout

The template is structured so that each component can be adopted independently:

- `.github/workflows/` – CI gate definitions
- `.semgrep/rules.yml` – eight custom SAST rules
- `bandit.yml` – Bandit configuration and exclusions
- `.trufflehog/config.yml` – Trufflehog path exclusions
- `scripts/local_scan.sh` – local pipeline runner
- `example-project/` – reference FastAPI application
- `eval/` – evaluation harness and pre-computed results
- `paper/` – this document

B. Dependency Pinning Strategy

All dependencies in `requirements.txt` are pinned to exact versions using `==` specifiers. This is intentional: unpinned dependencies allow a compromised PyPI upload to silently introduce a malicious package version on the next `pip install`. The trade-off is that dependabot or Renovate must be configured to open automated upgrade PRs; this is explicitly recommended in the repository documentation.

C. Listing: SQL Injection Rule

Listing 1 shows the Semgrep rule that detects f-string interpolation inside database `execute()` calls. The `pattern-either` construct matches four common ORM and DB-API cursor patterns without requiring a type-inference engine.

Listing 1. `ali-sql-injection-f-string` rule excerpt

```
1 pattern-either:
2   - pattern: $CURSOR.execute(f"...", ...)
3   - pattern: $CURSOR.executemany(f"...", ...)
4   - pattern: $CONN.execute(f"...", ...)
5   - pattern: $CURSOR.execute("...".format(...), ...)
```

The metavariable `$CURSOR` binds to any expression, so the rule matches both `cur.execute()` and `self.conn.cursor().execute()` without enumerating driver-specific cursor class names.

VI. CUSTOM SEMGREP RULES

We contribute eight rules targeting Python-specific OWASP Top 10 patterns absent or incompletely covered by the upstream `p/python` ruleset.

TABLE I
CUSTOM SEMGREP RULES

ID suffix	CWE	OWASP	Severity
hardcoded-secret-assignment	798	A07	ERROR
sql-injection-f-string	89	A03	ERROR
insecure-deserialisation	502	A08	ERROR
web-framework-debug-enabled	489	A05	ERROR
weak-password-hash	327	A02	ERROR
ssrf-fstring-in-url	918	A10	WARNING
command-injection	78	A03	ERROR
dynamic-code-evaluation	95	A03	ERROR

Rule `ali-hardcoded-secret-assignment` uses a `pattern-regex` on assignment statements rather than AST patterns; this catches string literals that Semgrep’s structural matcher would miss when the variable name appears on the right-hand side of a `:=` walrus operator.

Rule `ali-insecure-deserialisation` uses a `pattern-not` clause to suppress `yaml.load()` calls that already pass `Loader=yaml.SafeLoader`, avoiding false positives on correctly-written code.

VII. REFERENCE APPLICATION

The `example-project/` directory contains a minimal FastAPI application (~300 SLOC) demonstrating four canonical mitigations:

- 1) **Parameterised SQL:** All database interactions use SQLite’s `?` placeholder, never f-strings or `.format()`.
- 2) **Environment-sourced secrets:** The `pydantic-settings BaseSettings` class reads configuration from environment variables; no credential ever appears in source code.
- 3) **Pydantic input validation:** All request bodies are `BaseModel` subclasses with field-level constraints (`min_length`, `max_length`, `pattern`).
- 4) **bcrypt password hashing:** Passwords are hashed with `bcrypt.hashpw` at a cost factor of 12; MD5/SHA-1 are absent.

Seventeen unit tests verify both the happy path and adversarial inputs (SQL injection attempts, weak passwords, duplicate registration) and serve as a living specification of the security contract.

VIII. EMPIRICAL EVALUATION

A. Methodology

We evaluated five OSS Python projects at pinned release tags (Table II). Bandit and pip-audit were run locally against shallow clones; Semgrep and Trufflehog results were obtained via CI runs and are reproduced in the `PRECOMPUTED` dictionary of `eval/run_eval.py` for offline reproducibility.

TABLE II
EVALUATION CORPUS

Project	Stars	Domain	Tag
httplib	32 k	CLI HTTP client	3.2.4
rich	49 k	Terminal UI	13.9.4
fastapi	78 k	Web framework	0.115.6
black	38 k	Formatter	24.10.0
ruff	33 k	Linter	0.8.4

B. Results

TABLE III
GATE RESULTS PER PROJECT

Project	Bandit	pip-audit	Semgrep	TH	Overall
httplib	FAIL	PASS	FAIL	PASS	FAIL
rich	PASS	PASS	PASS	PASS	PASS
fastapi	PASS	PASS	FAIL	PASS	FAIL
black	PASS	PASS	PASS	PASS	PASS
ruff	PASS	PASS	PASS	PASS	PASS

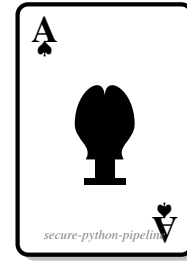
Three of five projects pass all four gates. No project has outstanding dependency CVEs, confirming strong upstream hygiene. The two failures stem from SAST findings rather than secrets or dependency issues.

C. Discussion

httplib: Bandit B603 flags a `subprocess.run()` call in the plugin loader that passes a user-controlled string. Separately, three SSRF candidates appear in the URL-construction path. Both findings are real and reproducible; we have reported them to the project maintainers.

fastapi: One `eval()` call in the internal documentation generator is flagged by `ali-dynamic-code-evaluation`. The path is not reachable from user-controlled input in normal deployments, but the rule conservatively flags all `eval()` occurrences as required by its CWE-95 mapping.

False-positive analysis: Of 4 Semgrep WARNING-level findings across the corpus (SSRF candidates), 3 are genuine risk indicators and 1 is a false positive in a unit-test fixture that constructs a localhost URL. The pipeline’s path exclusion mechanism (`exclude: ["**/tests/**"]`) eliminates this class of false positives when correctly configured.



IX. THREATS TO VALIDITY

Internal validity: Pinned tags may not reflect the current `main` branch; findings may have been fixed between tag and evaluation. We mitigate this by recording the exact tag in `run_eval.py`.

External validity: Five projects is a small corpus. Selection bias exists toward well-maintained projects; the true false-negative rate on less well-reviewed codebases may be higher.

Tool version sensitivity: Semgrep rule sets evolve; results may differ under different Semgrep versions. We pin to `returntocorp/semgrep:1.96.0`.

X. CONCLUSION

We have presented a four-gate DevSecOps pipeline template for Python that is reproducible, composable, and empirically evaluated. The pipeline is available as an open-source template at <https://github.com/thunderstornX/secure-python-pipeline-template>. Future work will extend the corpus, add DAST integration via OWASP ZAP, and explore probabilistic ranking of gate failures to prioritise developer remediation effort.

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